Zheng Xu

Department of Mathematics
Southern University of Science and Technology
Shenzhen 518055, China

(+86) 188-1179-6763
xuz6@sustech.edu.cn

Statement

I study algebraic geometry and I am a tenure-track assistant professor at the Southern University of Science and Technology(SusTech). I got my Ph.D. from Academy of Mathematics and Systems Science, Chinese Academy of Sciences (CAS) in June 2024, supervised by Professor Wenhao Ou. After that, I was a postdoctoral fellow of BICMR, Peking University, mentored by Professor Zhiyu Tian. My interests include birational geometry and Hodge theory in any characteristic.

Education

Ph.D., fundamental mathematics, Chinese Academy of Sciences (CAS), 2024

M.S., fundamental mathematics, Chinese Academy of Sciences (CAS), 2021

B.S., mathematics and applied mathematics, Hunan University of Science and Technology, 2018

Academic Employment

Southern University of Science and Technology, Shenzhen, China
Assistant Professor, 2026 - now

Peking University, Beijing, China
Postdoc, 2024 - 2026

Visiting positions

Visitor, Princeton University, May 3rd 2026 to June 13th 2026.

Visitor, Princeton University, April 13th 2025 to May 3rd 2025.

Visiting Student Research Collaborator, Princeton University, February 2024 to April 2024.

Papers

1. Emre Alp Özavcı, Zsolt Patakfalvi, Kevin Tucker, Joe Waldron, Zheng Xu. On rational chain connectedness of globally $+$-regular varieties. Submitted for a volume of the Summer Research Institute in Algebraic Geometry held at Colorado State University in 2025.

2. Jihao Liu, Zheng Xu. Non-algebraicity of non-abundant foliations and abundance for adjoint foliated structures. arXiv preprint arXiv:2510.04419, 2025.

3. Niklas Müller, Zheng Xu. Lagrangian Fibrations onto Varieties with Isolated Quotient Singularities. Journal für die reine und angewandte Mathematik (Crelles Journal). 832:297-307, 2026.

4. Jihao Liu, Zheng Xu. Non-vanishing implies numerical dimension one abundance. arXiv preprint arXiv:2505.05250, 2025.

5. Zheng Xu. Abundance for threefolds in positive characteristic when $\nu = 2$[J]. To appear in Compositio Mathematica.

6. Zheng Xu. NOTE ON THE THREE-DIMENSIONAL LOG CANONICAL ABUNDANCE IN CHARACTERISTIC >3. Nagoya Mathematical Journal. 255:694–723, 2024.

Invited Talks

Workshop on AI and AG, East China Normal University, July 2026

Algebraic Geometry Seminar, Northwestern University, May 2026

Workshop on Algebraic Geometry and Complex Geometry in Osaka, the University of Osaka, January 2026.

Workshop on Fano Varieties, Peking University, BICMR, June 2025

BICMR postdoc Seminar, Peking University, BICMR, June 2025

Algebraic Geometry Seminar, Tsinghua University, June 2025

Algebraic and Complex Geometry Symposium, East China Normal University, April 2025

Shandong Mathematical Society 2024 Annual Academy Meeting, Qingdao , January 2025

Algebraic Geometry Seminar, Peking University, BICMR, September 2024

Algebraic Geometry Seminar, Princeton University, Dept. of Mathematics, March 2024

Young Perspectives on Algebraic Geometry, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, December 2023

Southeast Algebraic Geometry Symposium, Southern University of Science and Technology, Dept. of Mathematics, April 2023

Beijing Guangzhou Algebraic Geometry Symposium, Online, January 2023

Language

Native Chinese (Mandarin)

Fluent English

August 3, 2026